

Video Transcript

Video 2: Overview of Hadoop: A Powerful and Extensible Platform (08:16)

Hadoop is one of those projects that changes the course of computing, especially when it comes to the data side of computing. The early days previous to the web were days in where, when it came to data, there was a lot less complexity. A lot of the organizations in the enterprises had data that was pretty well-structured in its generation and so storage fit neatly into rows and columns mostly as you can see. And so you had a world where you had a lot less velocity, variety, and volume. Now after the world goes through the revolution of the web, you certainly have a tremendous amount of variety.

The two way channel that is the web, allowed anyone to be able to produce data and certainly to intake data as well. So, that is an explosion. You have much more variety, you have velocity, the amount of data that is being produced is certainly overwhelming and you have a tremendous amount of volume as well. So, all of that doesn't fit neatly anymore into that rows and column, neat picture of the past and we really entered the era of big data that has very different needs. Now a company that was looking at this at the time was Google.

They were faced with the challenge of being able to parse the information that existed in the web, certainly a great deal of variety of velocity and volume. And so, the machines and the systems and the platforms that existed to be able to address this need simply didn't stretch to that. And so they needed to come up with a different way of doing this and they wanted to leverage commodity machines that they could simply throw all of this data into and from that a programming model that allowed you to be able to process those great deal big volumes of data.

And so they introduced a number of innovations into the market. And especially in this case, I want to emphasize Google File System, the fabric that stores all of that information, MapReduce, the programming model that allows you to process all of that data. And they added additional layers that are not as relevant to what we will be discussing here today when it comes to Hadoop. So, Hadoop tries to take those ideas. The software that Google had built was proprietary, but they shared their innovations and their ideas in carefully detailed publications.

One for the Google File system and other one for MapReduce, they did another one for Big Table, another layer. And, these were then used and they inspired a new generation of software engineers and data scientists and data engineers to build this new platform based on those ideas. And that idea is called Hadoop. And as you can see here, the very first layer of it was the distributed file system, which is the equivalent to Google's file system, GFS in this case called HDFS, to be able to store those volumes of data. And as you can see here, the fabric is responsible for taking those inputs of data, breaking them down into blocks and mapping them onto machines.

As you can see here, there is that varied or rich or unstructured data that then gets mapped onto those machines. And you can see there the block size that the default there being 128 megabytes and how that would be broken down. Now, it's worth noting that this was also or part of this platform was also the redundancy that was needed to be able to afford those failures in that commodity hardware. So, you can see there, the computing fabric was handling this. And in fact, one of the number that became well known was that there would be three copies made in case one of those failed.

And it turned out to be a good one, and one that has held for a long time now. The next layer was called the exact same thing as Google's, which was MapReduce, and that is, that model to be able to carry out computation. And this is one that existed even, or the idea of it existed even prior to Google, that you have a list of data onto which you're going to map functions, and then the output of those functions of each of the items of that data are going to be reduced onto a result.

And so here is a practical example of how you might do that. And in this case, this one's specific to Hadoop. And you can see there that input file from the famous book, it was the best of times, it was the worst of times, and so on. And how that could be a split there onto that list. Then, you map the functions onto it. And in this case, we're counting the frequency of words. And you can see there the values of the output of the mapping of the functions. Then, we would do a sort, and finally a reduction where we simply add up all the counts of those words, and you can see there the final result.

Now, this turns out to be the foundation upon which a great deal more is built. But even at this level, you can see here that you have a tremendously scalable resource that the world really didn't have and it was free, it was open-source, and it was shared openly. The last layer there is

one that I didn't mention, the YARN, yet another resource negotiator, and you can think of that as the management for this platform. So, what is it? More concisely, it's a scalable platform for distributed storage and distributed processing. And the next part of the definition there is the important one, that it is made for very large datasets. Really, this has no limitations.

You can simply throw hardware at it as your needs increase. And this scales horizontally, meaning that you simply need to add more machines which is a great thing. Now the other thing to note as well is that this was a big shift in how things had been done up until that point. The platforms that organizations had were very focused at a few specific use cases. The data warehouses at the time certainly were that. And here, you have a very different model, one that allows you to act upon tremendously large datasets. And as you can see there in the second block, increased computing power, fault tolerance, flexibility in data management.

That is, you could come up with any type of application that you wanted beyond the cookie-cutter solutions that existed in the data warehousing days at those times. And you had now a tremendous amount of resources that you could pick off the shelf that didn't require you to buy a license, that were open-source, and we see that shift that takes place in the industry. Nowadays, that is commonplace, that the biggest companies in the world, companies like Facebook, Amazon, many others are powered by open-source. But in those days, you started to see one of the first chefs.

You have this tremendous new platform that is robust, that is reliable, that is being used heavily in industry. And so, as you can see here, that becomes part of the story when it comes to big data needs in a platform to address them. You can still see here that even though this has been extended to capture a lot of the packages that are now new and that have emerged since then, you still have that footprint, you have HDFS, you have MapReduce, you have YARN, and then you have a host of other ones that are using those building blocks. But through it, you now have data storage, data processing, access and management. All of it in that new open-source paradigm. All of it addressing the multitude of needs when it comes to big data and big data needs.